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IRuDD: A Large-Scale Real-World Dataset for Industrial Rubber Defect Detection

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***Abstract*—Object detection, driven by advances in deep learning, has become a fundamental component of modern computer vision systems and serves as the core technology behind many industrial inspection tasks, including surface defect detection. In the rubber manufacturing industry, reliable quality control is critical not only for ensuring product safety and customer satisfaction, but also for improving production efficiency and reducing material waste. Despite its practical importance, large-scale real-world datasets for rubber surface inspection remain scarce. In this paper, we introduce IRuDD, a large-scale Industrial Rubber Defect Detection dataset comprising 20,828 high-resolution RGB images collected directly from production lines in real manufacturing environments. All images contain genuine surface defects arising during the production process and are annotated with bounding boxes for supervised defect localization, provided in both YOLO and COCO formats. Notably, the dataset does not rely on synthetic defects or artificial pre-processing, offering a realistic benchmark that reflects real-world conditions. To establish strong baselines and demonstrate the practical value of IRuDD, we conduct extensive experiments using a range of state-of-the-art object detection models, including CNN-based, YOLO-based, and transformer-based detectors, and report comprehensive quantitative results. By focusing on supervised defect detection under realistic constraints, IRuDD fills an important gap in industrial vision research and provides a solid foundation for future work on rubber surface inspection. The complete IRuDD dataset and benchmark code will be made publicly available upon acceptance at: https://github.com/arashlagzian/IRuDD.**

***Keywords*—rubber surface defect detection, surface defect detection, small object detection**

I. Introduction

The quality of manufactured products is significantly influenced by conditions during the production process. Surface defects, in particular, have a direct and often detrimental impact on product reliability, safety, and overall quality. In recent years, deep learning-based approaches have become the dominant paradigm for industrial surface defect detection. However, one of the primary challenges in this domain remains the limited availability of labeled data, which restricts model performance and generalization [1], [2].

To enable effective training of defect detection models, the availability of large-scale and well-annotated datasets is essential. Since the performance of modern object detection systems heavily depends on the quality and diversity of training data, numerous datasets have been proposed for different types of industrial surfaces [3]–[8]. These datasets cover a wide range of materials, including wood, fabric, plastic, metal, ceramics, composite materials, solar panels, and chip surfaces. Despite these advances, certain industrial domains remain underrepresented.

Rubber is a highly elastic material with unique mechanical and chemical properties, including flexibility, thermal resistance, and durability. Due to these characteristics, rubber products are widely used across various sectors, including transportation, manufacturing, and daily consumer applications. Ensuring the quality of rubber surfaces is therefore critical in industrial production pipelines. However, compared to other materials, there is a lack of large-scale, real-world datasets specifically designed for rubber surface defect detection.

To address this gap, we propose the Industrial Rubber Defect Detection dataset (IRuDD), a large-scale dataset collected directly from real-world production lines. The IRuDD dataset consists of 20,828 high-resolution images of rubber surfaces, each containing annotated defect regions with bounding boxes. The dataset captures a wide variety of real defects under practical conditions, including variations in lighting, viewpoint, and defect appearance. To ensure comprehensive coverage, images were collected from multiple angles and carefully annotated to localize all visible defect regions.

In addition to dataset construction, we conduct extensive experiments using multiple state-of-the-art object detection models to establish strong baselines on IRuDD. These experiments provide insights into the challenges of real-world defect detection, particularly in scenarios involving small and densely distributed anomalies.

The key novelties and contributions of this work are as

follows:

- We introduce IRuDD, the first large-scale real-world dataset specifically designed for industrial rubber surface defect detection, comprising 20,828 high-resolution images collected directly from active production lines — filling a critical gap absent in existing industrial vision benchmarks.
- We provide high-quality bounding box annotations in both YOLO and COCO formats, with 87.5% of defects classified as small objects, making IRuDD a uniquely challenging benchmark for fine-grained defect localization.
- We present the first comprehensive benchmark of state-of-the-art detection models on rubber surface data, covering CNN-based, YOLO-based, and transformer-based architectures under unified experimental conditions.
- We analyze the impact of real-world challenges including small object size and dense anomaly distribution, providing insights directly applicable to industrial deployment.

II. Related Works

A. Surface Defect Detection Methods

Surface defect detection has been widely studied across various industrial domains, including wood, fabric, plastic, metal, ceramics, composite materials, and road surfaces. Deep learning-based approaches, particularly convolutional neural networks (CNNs), have become the dominant methodology for these tasks.

In wood surface inspection, R-CNN-based methods and residual networks have been applied for defect detection, achieving promising performance through transfer learning and attention mechanisms [3], [9]–[12]. Similarly, in fabric inspection, advanced architectures such as cascaded Faster R-CNN and lightweight CNN models have been proposed to balance detection accuracy and computational efficiency [13]–[19].

For plastic and metal surfaces, YOLO-based methods and autoencoder-based approaches have been widely adopted due to their real-time performance and robustness [5], [20]–[23]. In ceramic and composite materials, multi-scale detection strategies and transformer-based models have demonstrated improved performance in detecting complex defect patterns [6], [24]–[29].

Additionally, defect detection in road and irregular surfaces, such as solar panels and chip manufacturing, has been addressed using transfer learning, YOLO variants, and attention-based models, with a particular focus on detecting small and sparse anomalies [7], [8], [30]–[37].

Despite the success of these approaches, most existing methods rely on datasets collected under controlled conditions or focus on specific materials, limiting their generalization to real-world industrial environments.

B. Industrial Defect Detection Datasets

The performance of defect detection models is strongly dependent on the availability of high-quality datasets. Several datasets have been introduced for different industrial applications, covering a wide range of materials and defect types. These datasets vary in scale, resolution, and annotation quality, and are often tailored to specific domains.

TABLE I
COMMON INDUSTRIAL PRODUCT SURFACE DEFECT DETECTION DATASETS. NS: NORMAL SAMPLES, DS: DEFECT SAMPLES, ALL: TOTAL NUMBER OF SAMPLES, AD: ANOMALY DETECTION, DET: OBJECT DETECTION, L: LOCALIZATION.

Dataset	NS	DS	All	Types of Objects	Resolution	Task Type
MVTec-AD [38]	4,096	1,258	5,354	15	700×1024	AD, L
BTAD [39]	952	392	1,344	3	600×1,600	AD, L
SSGD [40]	0	2,504	2,504	1	1500×1000	Det, L
VisA [41]	9,621	1,200	10,821	12	960×1562	AD, L
Real-IAD [40]	9,721	5,329	15,050	30	2000×5000	AD, L
Qrsted [28]	0	2,637	2,637	1	6720×4480	Det, L
DTU NordTank [42]	0	701	701	1	high-res	Det, L
IRuDD (ours)	0	20,828	20,828	1	960×1280	Det, L

However, most existing datasets either contain a limited number of samples, rely on synthetic defects, or do not adequately represent real-world production conditions. In particular, datasets for rubber surface defect detection are largely absent, creating a gap between research and industrial application.

C. Research Gap

Although significant progress has been made in both detection algorithms and dataset construction, IRuDD differs from existing benchmarks in several key aspects beyond scale. First, it is the only dataset targeting rubber surfaces specifically, whereas existing datasets cover ceramics, metals, or composite materials. Second, all images are collected directly from active production lines without synthetic augmentation or lab-controlled conditions, ensuring genuine real-world variability in lighting, viewpoint, and defect appearance. Third, unlike anomaly detection datasets such as MVTec-AD and VisA which require normal samples, IRuDD is designed exclusively for supervised defect localization under realistic industrial constraints. Finally, with 87.5% of instances classified as small objects, IRuDD presents a distinctly challenging benchmark that existing datasets do not adequately represent.

III. IRuDD Dataset

A. Dataset Overview

The IRuDD dataset is a large-scale collection of real-world rubber surface images designed for defect detection tasks. A total of 20,828 high-resolution images were selected from approximately 26,000 captured samples. Each image contains at least one defect instance, with up to 16 defects per image.

In this work, all surface irregularities are treated as a single defect class, defined as any visible deviation from a normal rubber surface. This includes, but is not limited to, cavities, bumps, scratches, burns, and color variations, as illustrated in Fig. 2. This unified labeling strategy is intentional — it prioritizes robust defect localization under real industrial conditions over fine-grained classification. Per-category annotation and balance analysis are planned for future versions of the dataset. All defect instances are annotated with bounding boxes and provided in both YOLO and COCO formats to support a wide range of detection frameworks.

Compared to existing datasets (Table I), IRuDD provides a larger number of defect samples collected under real industrial conditions, making it a challenging benchmark for practical applications.

B. Data Acquisition Setup

To ensure consistent and high-quality image acquisition, we designed a custom data collection device equipped with an industrial camera and controlled lighting sources. The camera was mounted approximately 25 cm above the rubber surface, enabling full coverage of the inspected area.

A 5 MP Imaging Source industrial camera (5 mm fixed lens, 960×1280 px output) was used to capture detailed surface textures. This resolution was selected to balance image detail with real-time processing requirements on the production line conveyor system. Multiple lighting configurations were evaluated, and the optimal setup was selected to minimize noise and enhance defect visibility. Images were collected over a period of three months across two production lines, covering six rubber product types to ensure diversity in surface appearance and defect distribution. Rubber products were transported through the system using a conveyor mechanism, ensuring consistent positioning and acquisition conditions.

C. Annotation Process

The dataset was annotated using Label Studio in object detection mode. To ensure annotation quality, three annotators labeled the data sequentially, followed by a final verification step. This multi-stage process reduced labeling errors and improved consistency.

Annotated samples are illustrated in Fig. 1, where defect regions are highlighted and magnified for clarity.

D. Dataset Statistics

The statistical properties of IRuDD are summarized in Table II. The dataset contains 66,492 annotated defect instances, with an average of 3.2 objects per image. The mean bounding box size is 16.77×31.38 px (median area: 298.19 px^2), with 87.5% of defects classified as small ($\leq$1,024 px^2), 11.3% as medium, and only 1.2% as large, based on COCO size thresholds. This distribution confirms the challenging nature of IRuDD as a small-object detection benchmark, where defect regions are localized and sparsely distributed across the image surface.

Representative samples of defects are shown in Fig. 2.

E. Data Preparation for Training

Before model training, standard preprocessing techniques were applied. Images were normalized to ensure consistent input distributions, and data augmentation techniques such as flipping, rotation, and noise injection were used to improve model robustness.

TABLE II
STATISTICAL PROPERTIES OF THE IRUDD DATASET.

Property	**IRuDD**
Number of images	20,828
Number of objects	66,492
Avg. objects per image	3.2
Max objects per image	16
Min objects per image	1
Total defect area	52 M (px^2)
Avg. defect area	2,507.2 (px^2)
Largest defect	32,943.8 (px^2)
Smallest defect	1.16 (px^2)
Defect-to-surface ratio	0.2
Avg. bbox width (px)	16.77
Avg. bbox height (px)	31.38
Median defect area (px^2)	298.19
Small defects ($\leq$1,024 px^2)	58,167 (87.5%)
Medium defects (1,024–9,216 px^2)	7,500 (11.3%)
Large defects ($>$9,216 px^2)	825 (1.2%)

IV. EXPERIMENTAL SETUP

A. Detection Models

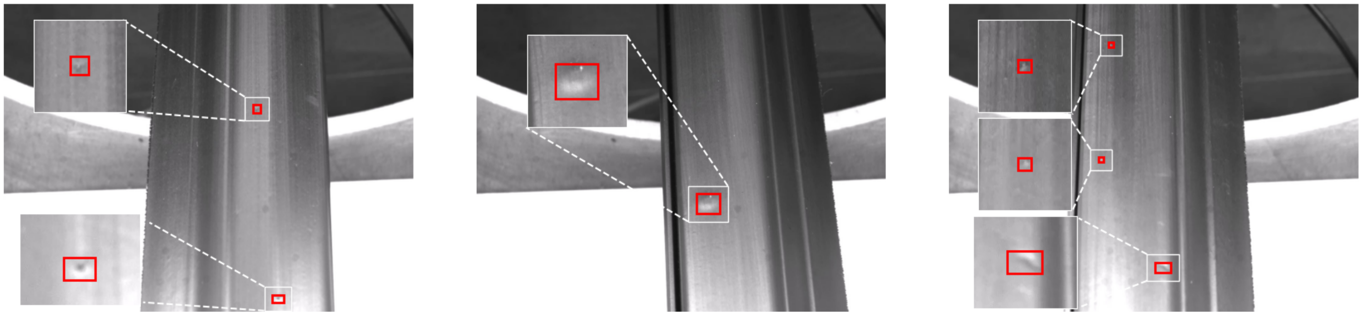
To evaluate the effectiveness of the IRuDD dataset, we benchmark several state-of-the-art object detection models, including both two-stage and one-stage detectors. Specifically, we evaluate Faster R-CNN [43], YOLOv5, YOLOv7, YOLOv8, YOLOv10, RT-DETR [44], and HIC-YOLOv5 [45]. These models represent a diverse set of architectures, including CNN-based and transformer-based detectors. HIC-YOLOv5 is an improved variant of YOLOv5 specifically designed for small object detection, making it a particularly relevant baseline given the prevalence of small defect regions in IRuDD.

B. Training Configuration

All models are implemented using the PyTorch framework and trained under a unified experimental setup on an NVIDIA RTX 3060 8GB GPU. The dataset is randomly split into training, validation, and test sets using a fixed seed (seed = 42), following an 80/10/10 ratio, resulting in 16,662 training images, 2,083 validation images, and 2,083 test images.

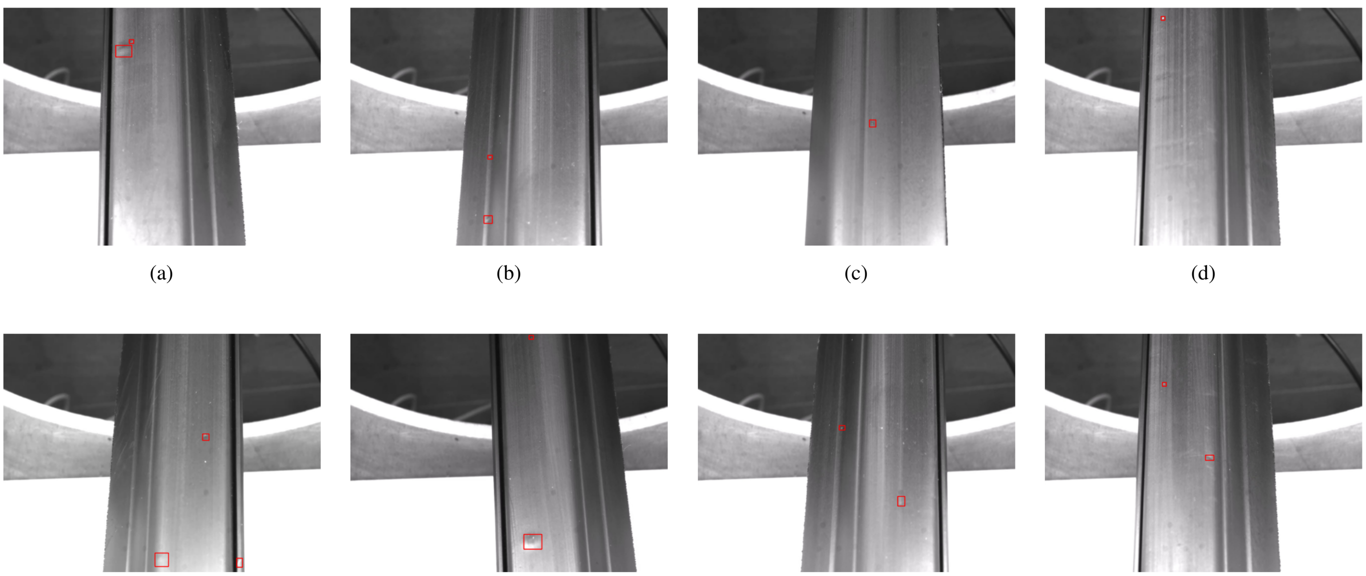
All models are initialized with COCO pretrained weights and fine-tuned on IRuDD. Two-stage detectors (Faster R-CNN) receive full-resolution inputs of 960×1280 px with a learning rate of 0.005, while YOLO-based and transformer-based models use resized inputs of 640×640 px with their respective default learning rates. All models are trained with a batch size of 4, except RT-DETR which is additionally evaluated with a batch size of 8. Training runs for 200 epochs without early stopping; the checkpoint with the best validation performance is saved and used for evaluation.

Data augmentation techniques are applied to improve model generalization, including horizontal flipping, color jittering (brightness, contrast, saturation, and hue set to 0.2), Gaussian noise injection with variance in the range of (10.0, 50.0), and uniform blur with a kernel size of 3. Augmentations are applied with a probability of 0.5. Due to the fixed camera



(a) (b) (c)

Fig. 1. Example annotations with magnified defect regions.



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 2. Representative defect samples (a-h) from the IRuDD dataset, illustrating the visual diversity of surface anomalies including cavities, scratches, burns, and color variations. All instances are annotated under a unified defect class.

setup, geometric transformations such as scaling and zooming are not applied.

To ensure reproducibility, all models are based on publicly available implementations, trained with a fixed random seed (seed = 42), and the complete IRuDD dataset will be publicly released upon acceptance, allowing the community to verify and reproduce all reported results.

C. Evaluation Metrics

We evaluate model performance using standard object detection metrics, including Precision, Recall, and mean Average Precision (mAP). In particular, we report mAP@50 and mAP@50–90 to assess both coarse and fine localization performance.

V. Results and Analysis

A. Benchmark Results

The performance of different detection models on the IRuDD dataset is summarized in Table III. Among all evaluated methods, YOLO-based detectors achieve the best performance in terms of Precision, Recall, and mAP@50, with YOLOv5 obtaining the highest mAP@50 score of 98.00. However, Faster R-CNN with VGG-16 achieves the highest mAP@50–90 score of 76.10, suggesting stronger fine-grained localization at stricter IoU thresholds, likely due to the two-stage refinement mechanism of region proposal networks. YOLOv8 and YOLOv10 also demonstrate strong performance, indicating the effectiveness of one-stage detectors for this task.

In contrast, transformer-based models such as RT-DETR show comparatively lower performance, particularly in mAP@50–90, suggesting challenges in precise localization of small defect regions.

TABLE III
PERFORMANCE METRICS FOR DIFFERENT MODELS ON THE IRUDD DATASET.

Model	Backbone	Precision	Recall	mAP@50	mAP@50–90
Faster-RCNN	ResNet-50	69.15	83.62	61.05	63.08
Faster-RCNN	VGG-16	74.81	80.51	74.81	**76.10**
Faster-RCNN	MobileNetV3	53.96	65.34	58.65	57.40
YOLOv5	CSPDarknet53	**94.10**	**96.00**	**98.00**	65.50
YOLOv7	E-ELAN	64.30	65.20	69.80	35.60
YOLOv8	Custom-CSPDarknet53	92.10	91.20	96.50	64.10
YOLOv10	CSPNet	89.30	85.60	93.50	60.30
RT-DETR (b=4)	ResNet	57.90	47.60	56.20	23.90
RT-DETR (b=8)	ResNet	64.70	55.90	65.90	28.60
HIC-YOLOv5	CSP	56.14	58.20	57.72	27.05

TABLE IV
PERFORMANCE METRICS FOR DIFFERENT MODELS WITH DATA AUGMENTATION.

Model	Backbone	Precision	Recall	mAP@50	mAP@50–90
Faster-RCNN	MobileNetV3-s	39.17	62.68	44.41	43.84
YOLOv5	CSP-Darknet53	**95.20**	**96.80**	**98.10**	67.40
YOLOv7	E-ELAN	82.40	78.80	87.90	52.80
YOLOv8	Custom-CSPDarknet53	92.60	93.00	96.70	65.60
YOLOv10	CSPNet	92.60	88.60	95.50	65.70

Since all evaluated models are unmodified open-source implementations, inference speed is hardware-dependent and consistent with benchmarks reported in their respective original publications [43]–[45]. As this work focuses on dataset contribution rather than real-time deployment, detailed speed profiling is beyond the scope of this benchmark.

B. Effect of Data Augmentation

The impact of data augmentation is presented in Table IV. We observe consistent improvements across YOLO-based models, with YOLOv5 achieving an mAP@50 of 98.10. This indicates that augmentation enhances robustness to variations in lighting and noise, which are common in real-world industrial environments.

C. Analysis of Model Behavior

YOLO-based detectors consistently outperform two-stage detectors such as Faster R-CNN in terms of mAP@50. This can be attributed to the dense prediction mechanism of one-stage detectors, which is more effective for capturing small and densely distributed defect regions.

Transformer-based models such as RT-DETR exhibit lower performance, likely due to their reliance on global context, which is less informative for small, localized anomalies.

Overall, the results highlight the importance of multi-scale feature representation and dense detection strategies for real-world industrial defect detection tasks.

VI. LIMITATION

IRuDD has several limitations: it contains only defective samples without normal images, restricting its use in anomaly detection settings; all defects are grouped into a single class, limiting fine-grained classification; and the controlled acquisition setup may reduce generalization to other environments. Additionally, the dataset is specific to rubber surfaces, and defect regions are often small, posing challenges for detection models. Future work will focus on incorporating defect-free samples, introducing multi-class defect labels, and extending coverage to broader rubber product types and manufacturing conditions.

VII. CONCLUSION

In this paper, we introduced IRuDD, a large-scale real-world dataset for industrial rubber defect detection, consisting of 20,828 high-resolution RGB images collected directly from production lines. The dataset provides comprehensive annotations for object detection tasks and reflects realistic industrial conditions, including variations in defect appearance and distribution.

We conducted extensive experiments using multiple state-of-the-art object detection models to establish strong baselines on IRuDD. The results demonstrate that one-stage detectors, particularly YOLO-based models, achieve superior performance on this dataset, highlighting the importance of dense prediction mechanisms for detecting small and localized defects.

By addressing the lack of real-world datasets in rubber surface inspection, IRuDD serves as a challenging and practical benchmark for industrial defect detection research. The dataset bridges the gap between academic development and real-world deployment, supporting the design of robust and scalable inspection systems.

For future work, we plan to expand the dataset by incorporating additional samples with a higher proportion of small-scale defects and exploring more diverse acquisition conditions. We also aim to extend the dataset to support fine-grained defect classification and evaluate advanced detection frameworks under more complex industrial scenarios.

The IRuDD dataset will be publicly released upon acceptance of this paper, enabling reproducible research and fostering further advancements in industrial defect detection.

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